

nested for loops

CK 141 nested for loops

So what does this print?

```
int sum = 0;
for (int j = 0; j < 3; j++) {
    for (int i = 0; i < 10; i++) {
        sum += 1;
    }
}
System.out.println(sum);
```

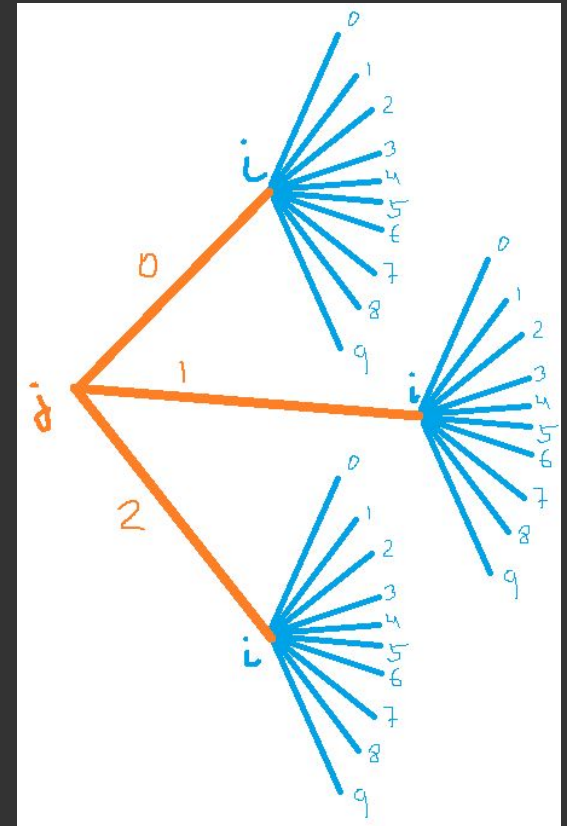
- A: 3
- B: 10
- C: 13
- D: 0
- E: 30

So what does this print?

```
int sum = 0;
for (int j = 0; j < 3; j++) {
    for (int i = 0; i < 10; i++) {
        sum += 1;
    }
}
System.out.println(sum);
```

- A: 3
- B: 10
- C: 13
- D: 0
- E: 30**

You can visualize this [here](#).



What does this print?

```
int outer = 0;
int inner = 0;
for (int j = 0; j < 100; j++) {
    outer++;
    for (int i = 0; i < 75; i++) {
        inner++;
    }
}
System.out.println(outer + ", " + inner);
```

A: 0, 0

B: 100, 75

C: 100, 7500

D: 7500, 750000

E: none of the above

What does this print?

```
int outer = 0;
int inner = 0;
for (int j = 0; j < 100; j++) {
    outer++;
    for (int i = 0; i < 75; i++) {
        inner++;
    }
}
System.out.println(outer + ", " + inner);
```

A: 0, 0

B: 100, 75

C: 100, 7500

D: 7500, 750000

E: none of the above

What does this print?

```
int inner1 = 0, inner2 = 0, outer = 0;
for (int i=0; i<6; i++) {
    outer++;
    for (int j=0; j<=3; j++) {
        inner1++;
    }
}
inner2++;
System.out.println("outer " + outer + ", inner1 " + inner1 +
    ", inner2 " + inner2);
```

- A: outer 6, inner1 3, inner2 18
- B: outer 6, inner1 4, inner2 1
- C: outer 7, inner1 3, inner2 18
- D: outer 7, inner1 24, inner2 18
- E: outer 6, inner1 24, inner2 1

What does this print?

```
int inner1 = 0, inner2 = 0, outer = 0;
for (int i=0; i<6; i++) {
    outer++;
    for (int j=0; j<=3; j++) {
        inner1++;
    }
}
inner2++;
System.out.println("outer " + outer + ", inner1 " + inner1 +
    ", inner2 " + inner2);
```

A: outer 6, inner1 3, inner2 18

B: outer 6, inner1 4, inner2 1

C: outer 7, inner1 3, inner2 18

D: outer 7, inner1 24, inner2 18

E: outer 6, inner1 24, inner2 1

What does this print?

```
int sum = 0;
for (int j = 0; j < 10; j++) {
    for (int i = 0; i < 75; i++) {
        for (int k = 0; k < 20; k++) {
            sum++;
        }
    }
}
System.out.println(sum);
```

A: 0

B: 7500

C: 15000

D: 10 75 20

E: none of the above

What does this print?

```
int sum = 0;
for (int j = 0; j < 10; j++) {
    for (int i = 0; i < 75; i++) {
        for (int k = 0; k < 20; k++) {
            sum++;
        }
    }
}
System.out.println(sum);
```

A: 0

B: 7500

C: 15000 //sum++; is executed $j*i*k$ times!

D: 10 75 20

E: none of the above

What are the values of outer, inner1, and inner2 (X, Y, Z are positive ints)

```
int outer = 0;
int inner1 = 0;
int inner2 = 0;
for (int i = 0; i < X; i++) {
    outer++;
    for (int j = 0; j < Y; j++) {
        inner1++;
    }
    for (int k = 0; k < Z; k++) {
        inner2++;
    }
}
```

Answer	outer	inner1	inner2
A	X	Y	Z
B	X	X*Y	X*Z
C	X	X*Y	X*Y*Z
D	X	X*Y	Z
E	X	Y	Y*Z

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int outer = 0;
int inner1 = 0;
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        inner2++;
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}
```

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    inner2++;
}
```

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    inner2++;
}
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    inner1++;
}
for (int k = 0; k < Z; k++) {
    inner2++;
}
```

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int outer = 0;
int inner1 = 0;
int inner2 = 0;
for (int i = 0; i < X; i++) {
    outer++;
    for (int j = 0; j < Y; j++) {
        inner1++;
        for (int k = 0; k < Z; k++) {
            inner2++;
        }
    }
}
```

Answer	outer	inner1	inner2
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D	X	X*Y	Z
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int inner2 = 0;
for (int i = 0; i < X; i++) {
    outer++;
    for (int j = 0; j < Y; j++) {
        inner1++;
        for (int k = 0; k < Z; k++) {
            inner2++;
        }
    }
}
```

Answer	outer	inner1	inner2
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    inner1++;
    for (int k = 0; k < Z; k++) {
        inner2++;
    }
}
```

Answer	outer	inner1	inner2
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int outer = 0;
int inner1 = 0;
int inner2 = 0;
for (int i = 0; i < X; i++) {
    outer++;
}
for (int j = 0; j < Y; j++) {
    inner1++;
    for (int k = 0; k < Z; k++) {
        inner2++;
    }
}
```

Answer	outer	inner1	inner2
A	X	Y	Z
B	X	X*Y	X*Z
C	X	X*Y	X*Y*Z
D	X	X*Y	Z
E	X	Y	Y*Z

What does this print?

```
int outer=0;
int inner=0;
for (int i=0; i<6; i++) {
    outer++;
    for (int j=0; j<=i; j++) {
        inner++;
    }
}
System.out.println("outer "+outer+", inner "+inner);
```

A: outer 6, inner 24

B: outer 7, inner 6

C: outer 7, inner 21

D: outer 6, inner 21

E: outer 6, inner 30

What does this print?

```
int outer=0;
int inner=0;
for (int i=0; i<6; i++) {
    outer++;
    for (int j=0; j<=i; j++) {
        inner++;
    }
}
System.out.println("outer "+outer+", inner "+inner);
```

A: outer 6, inner 24

B: outer 7, inner 6

C: outer 7, inner 21

D: outer 6, inner 21

E: outer 6, inner 30

What does this print?

```
int outer=0;
for (int i=0; i<12; i++) {
    if (i % 2 == 0) {
        outer++;
    }
}
System.out.println(outer);
```

A: 12

B: 6

C: 3

D: 1

E: 0

What does this print?

```
int outer=0;
for (int i=0; i<12; i++) {
    if (i % 2 == 0) {
        outer++;
    }
}
System.out.println(outer);
```

A: 12

B: 6

C: 3

D: 1

E: 0

testing even vs odd

$x \% 2 == 0$ is true when x is even

$x \% 2 == 1$ is true when x is odd

What does this print?

```
int outer=0;
for (int i=1; i<=24; i++) {
    if (i % 2 == 0 || i % 3 == 0) {
        outer++;
    }
}
System.out.println(outer);
```

A: 2

B: 4

C: 8

D: 16

E: 24

What does this print?

```
int outer=0;
for (int i=1; i<=24; i++) {
    if (i % 2 == 0 || i % 3 == 0) {
        outer++;
    }
}
System.out.println(outer);
```

A: 2

B: 4

C: 8

D: 16

E: 24

What does this print?

```
int outer=0;
for (int i=1; i<=10; i++) {
    if (i % 2 == 0) {
        outer++;
    } else {
        outer += 2;
    }
}
System.out.println(outer);
```

A: 5

B: 10

C: 15

D: 20

E: 9

What does this print?

```
int outer=0;
for (int i=1; i<=10; i++) {
    if (i % 2 == 0) {
        outer++;
    } else {
        outer += 2;
    }
}
System.out.println(outer);
```

A: 5

B: 10

C: 15

D: 20

E: 9

What does this print?

```
int outer=0;
for (int i=1; i<=10; i++) {
    if (i % 2 == 1) {
        outer++;
    } else {
        i++;
    }
}
System.out.println(outer);
```

A: 0

B: 1

C: 10

D: 5

E: 9

What does this print?

```
int outer=0;
for (int i=1; i<=10; i++) {
    if (i % 2 == 1) {
        outer++;
    } else {
        i++;
    }
}
System.out.println(outer);
```

A: 0

B: 1

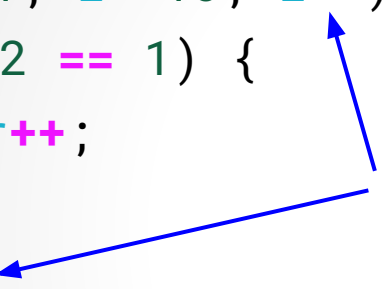
C: 10

D: 5

E: 9

Basically, as soon as we hit an even number, we jump ahead 2 spots

```
int outer=0;
for (int i=1; i<=10; i++) {
    if (i % 2 == 1) {
        outer++;
    } else {
        i++;
    }
}
```



These both add 1 to i!

So once we hit the else, variable i never hits another odd number!

Because we keep adding 2 to i, which gives us the next even number.

What does this print?

```
int outer=0;
for (int i=0; i<10; i++) {
    if (x != 0 && x != 9) {
        outer++;
    }
}
```

```
System.out.println(outer);
```

A: 0

B: 5

C: 8

D: 10

E: 15

What does this print?

```
int outer=0;
for (int i=0; i<10; i++) {
    if (x != 0 && x != 9) {
        outer++;
    }
}
System.out.println(outer);
```

A: 0

B: 5

C: 8

D: 10

E: 15

x must be either not 0 or not 9, since it cannot both be 0 and 9